



Solve with us 6.1 - Not Graded



This assignment will not be graded and is only for practice.

Key Points: 1

- A map $T: V \rightarrow W$ between two vector spaces V and W is said to be a linear transformation if for any two vectors v_1 and v_2 in the vector space V and for any scalar c in R the following conditions hold:
 - $T(v_1 + v_2) = T(v_1) + T(v_2)$
 - $T(cv_1) = cT(v_1)$

1 point

Let T be a function from R^2 to R^2 defined as follows:

$$T: R^2 \rightarrow R^2 \quad T(x, y) = (2x, y)$$

Choose the set of correct option(s).

[Hint: Assume $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$, then check the properties of linear transformation for the given function T .]

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$T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in R^2$.

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$T(cv_1) = cT(v_1)$ for all $v_1 \in R^2$, and $c \in R$.

☐

T is a linear transformation.

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T is not a linear transformation.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in R^2$.

$T(cv_1) = cT(v_1)$ for all $v_1 \in R^2$, and $c \in R$.

T is a linear transformation.

1 point

Let T be a function from R^2 to R^2 defined as follows:

$$T: R^2 \rightarrow R^2 \quad T(x, y) = (x - 1, y + 1)$$

Choose the set of correct option(s).

[Hint: Assume $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$, then check the properties of linear transformation for the given function T .]

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$T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in R^2$.

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$T(cv_1) = cT(v_1)$ for all $v_1 \in R^2$, and $c \in R$.

☐

T is a linear transformation.

☐

T is not a linear transformation.



No, the answer is incorrect.

Score: 0

Accepted Answers:

T is not a linear transformation.

Key Points: 2

- A linear transformation $T: V \rightarrow W$ is said to be an isomorphism if it is bijective, i.e. the followings hold:
 - $T(v_1) = T(v_2)$ implies $v_1 = v_2$ (Injectivity).
 - For any $w \in W$ there exists $v \in V$ such that $T(v) = w$ (Surjectivity).

1 point

If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$, then which of the following options is true?

[[Hints:

Hint 1: Let $v_1 = (x_1, y_1)$, $v_2 = (x_2, y_2)$, and assume that $T(v_1) = T(v_2)$. Then check whether $v_1 = v_2$ or not.

Hint 2: Does there exist any $v \in \mathbb{R}^2$ for which $T(v) = (0, 1)$?

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T is both one to one and onto.

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T is one to one but not onto.

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T is onto but not one to one.

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T is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is neither one to one nor onto.

1 point

If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation defined by

$T(x, y) = (2x + 3y, 5x - y, x + 6y)$. Which of the following options is true?

[Hint: Does there exist any $v \in \mathbb{R}^2$ for which $T(v) = (1, 0, 0)$?

☐

T is both one to one and onto.

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T is one to one but not onto.

☐

T is onto but not one to one.

☐

T is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is one to one but not onto.

Key Points: 3Key Points: 3

- Standard Ordered basis of R^n R^n :

$$\{e_i \mid i = 1, 2, \dots, n\}, \{e_i \mid i = 1, 2, \dots, n\},$$

where $e_1 = (1, 0, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, $e_3 = (0, 0, 1, \dots, 0)$, $\dots$, $e_n = (0, 0, \dots, 0, 1)$. In general, $e_i = (0, 0, \dots, 0, 1, 0, \dots, 0)$ where 1 is at the i -th position and 0 elsewhere.

- If $T(e_i)$ is known for all i , then the description of T can be known entirely. Suppose $T(e_i) = v_i$, for $i = 1, 2, \dots, n$, then we have

$$\begin{aligned} T(x_1, x_2, \dots, x_n) &= T(x_1 e_1 + x_2 e_2 + \dots + x_n e_n) = T(x_1 e_1 + x_2 e_2 + \dots + x_n e_n) \\ &= T(x_1 e_1) + T(x_2 e_2) + \dots + T(x_n e_n) = x_1 T(e_1) + x_2 T(e_2) + \dots + x_n T(e_n) = x_1 v_1 + x_2 v_2 + \dots + x_n v_n \\ &= T(x_1 e_1) + T(x_2 e_2) + \dots + T(x_n e_n) = x_1 T(e_1) + x_2 T(e_2) + \dots + x_n T(e_n) = x_1 v_1 + x_2 v_2 + \dots + x_n v_n \end{aligned}$$

1 point

Let $T: R^2 \rightarrow R^2$ be a linear transformation, such that $T(1, 0) = (3, 2)$ and $T(0, 1) = (1, 5)$. Which of the following options is true?

[Hint: $\{(1, 0), (0, 1)\}$ is the standard ordered basis of R^2 .]

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$$T(x, y) = (3x + y, 2x + 5y)$$

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$$T(x, y) = (3x + 2y, x + 5y)$$

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$$T(x, y) = (5x, 6y)$$

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$$T(x, y) = (4x, 7y)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(x, y) = (3x + y, 2x + 5y)$$

1 point

Let $T: R^2 \rightarrow R^2$ be a linear transformation, such that $T(2, 1) = (3, 2)$ and $T(1, 3) = (0, 0)$. Which of the following options is true?

[Hint: Express $(1, 0)$ and $(0, 1)$ in terms of $(2, 1)$ and $(1, 3)$.]

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$$T(x, y) = \frac{1}{5}(3x, 2y)$$

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$$T(x, y) = (x + 1, y + 1)$$

☐

$$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$$

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T cannot be determined from the given information.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

$$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y) \quad T(x, y) = 51(9x - 3y, 6x - 2y)$$

Key Points: 4Key Points: 4

- Let $T: V \rightarrow W$ be a linear transformation. Let $\beta = \{v_1, v_2, \dots, v_n\}$ be an ordered basis of V and $\gamma = \{w_1, w_2, \dots, w_m\}$ be an ordered basis of W . So each $f(v_i)$ can be uniquely written as linear combination of w_j 's, where $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

$$\begin{aligned} T(v_1) &= a_{11}w_1 + a_{21}w_2 + \dots + a_{m1}w_m & T(v_2) &= a_{12}w_1 + a_{22}w_2 + \dots + a_{m2}w_m & \vdots & T(v_n) = a_{1n}w_1 \\ T(v_1) &= a_{11}w_1 + a_{21}w_2 + \dots + a_{m1}w_m & T(v_2) &= a_{12}w_1 + a_{22}w_2 + \dots + a_{m2}w_m & \vdots & T(v_n) = a_{1n}w_1 \\ &+ a_{2n}w_2 + \dots + a_{mn}w_m \end{aligned}$$

The matrix corresponding to the linear transformation T with respect to the ordered bases β and γ is given by,

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

The above matrix not only depends on the linear map T , but also on the choice of the ordered bases β for V and γ for W .

- It is important to note that, for a fixed choice of basis of V and W , there is a one to one correspondence between the set of all linear transformations from V to W , and the set of all $m \times n$ matrices, where $n = \dim(V)$ and $m = \dim(W)$.

1 point

Suppose $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$. Which of the following matrices corresponds to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain?

[Hint: $T(1, 0) = (1, 0)$ and $T(0, 1) = (0, 0)$]

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$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

1 point

Suppose $T: R^2 \rightarrow R^2$ is a linear transformation, such that $T(x, y) = (x + y, x - y)$. Which of the following matrices corresponds to T with respect to the standard ordered basis of R^2 , i.e., $\{(1, 0), (0, 1)\}$, for the domain and $\{(1, 1), (1, -1)\}$ for the co-domain?

[Hint: $T(1, 0) = (1, 1)$ and $T(0, 1) = (1, -1)$]

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$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Feedback:

• $T(1, 0) = (1, 1) = 1(1, 1) + 0(1, -1)$, and $T(0, 1) = (1, -1) = 0(1, 1) + 1(1, -1)$.

Accepted Answers:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

1 point

Let $W = \{(x, y, z) \mid x = 2y + z\}$ be a subspace of R^3 . Let $\beta = \{(2, 1, 0), (1, 0, 1)\}$ be a basis of W . Let $T: W \rightarrow R^2$ be a linear transformation such that $T(2, 1, 0) = (1, 0)$ and $T(1, 0, 1) = (0, 1)$. What will be the matrix representation of T with respect to the basis β for W and $\gamma = \{(1, 1), (1, -1)\}$ for R^2 ?

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$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & -1 \end{bmatrix}$$

1 point

Let $\beta = \{(1, 0), (0, 1)\}$ and $\gamma = \{(1, 1), (1, -1)\}$ be two bases of R^2 . If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is the matrix representation of a linear transformation $T: R^2 \rightarrow R^2$ with respect to β for the both the domain and co-domain. What will be the matrix representation of T with respect to γ for the domain and β for the co-domain?

[Hint: First find $T(x, y)$ and then find $T(1, 1)$ and $T(1, -1)$.]

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$$\begin{bmatrix} a + b & c + d \\ a - b & c - d \end{bmatrix}$$

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$$\begin{bmatrix} a - b \\ c - d \end{bmatrix}$$

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$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix}$$

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$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Feedback:

• $T(x, y) = (ax+by, cx+dy)$, so $T(1, 1) = (a+b, c+d)$ and $T(1, -1) = (a-b, c-d)$.

Accepted Answers:

$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix}$$